

Emergency Response System for Organisations

Design, Prototype Evolution, Experimental Evaluation, and
Commercialisation Strategy

Prepared by:

Research & Development Engineering Team
Micromax Technology Pty Ltd

Project Lead & Mentorship:

Shakif Aziz
Research and Development Team Leader
Micromax Technology

Contributing Engineers:

Thomas Speer, Ziyue Li, Zachary Wark,
Nur Azmina Mohd Rashidi, Oliver Michels, Laila Salem

August 20, 2026

Executive Summary

This technical engineering report presents the design, prototype evolution, empirical performance analysis, and commercialisation strategy of the **Emergency Response System (ERS) for Organisations**. Developed by the Research and Development Engineering Division at Micromax Technology Pty Ltd, this platform addresses statutory Work Health and Safety (WHS) compliance mandates and duty-of-care obligations for lone, isolated, and high-risk workers across industrial, manufacturing, logistics, and healthcare environments.

Evolving from an initial academic proof-of-concept (ECTE351) to a fully functional, field-tested industrial prototype (ECTE399 R&D), the system bridges the gap between theoretical embedded design and robust on-premises deployment. By eliminating the recurring Software-as-a-Service (SaaS) subscription fees characteristic of commercial competitors, the ERS provides an economically viable, one-off capital expenditure (CAPEX) solution offering complete on-premises data ownership and direct facility hardware integration.

Key Technological Breakthroughs & Deliverables

- **Deterministic Multi-Modal Emergency Detection:** Engineered wearable client firmware capable of distinguishing deliberate Personal Emergencies (> 2.0 s press-and-hold) from facility-wide Mass Emergencies (5 rapid clicks in < 3.0 s), alongside solving underlying microcontroller GPIO pin-sharing reset conflicts.
- **Kinematic Fall Detection with Grace Window:** Implemented a 6-DOF inertial measurement algorithm capturing freefall ($< 0.5g$) and traumatic impact ($> 2.8g$) while rejecting routine daily motions (e.g., prayer prostration, stair descent), coupled with a 10-second acoustic/visual cancellation grace window that achieved a 100% false-alarm suppression rate.
- **Dual-Redundant Wireless Telemetry:** Developed a dual-link communication architecture prioritizing sub-GHz LoRaWAN (AU915, SF7) utilizing an ultra-compact 4-byte binary payload (36.1 ms airtime, 98.4% packet delivery rate through structural steel) with an autonomous 3.96 second failover to IEEE 802.11 Wi-Fi.
- **Symbolic Indoor Positioning via Room Zone Mapping:** Deployed a Bluetooth Low Energy (BLE) gateway infrastructure and established that Mamdani Fuzzy Inference mapping directly to polygon-based Room Zones achieves a 94.8% classification accuracy, resolving severe RF multipath anomalies inherent to steel-clad industrial buildings.
- **Centralized Edge Telemetry & Multi-Channel Dispatch:** Deployed an on-premises Raspberry Pi 5 server hosting multithreaded ingress listeners, an SQLite audit database, a web-based floor plan zoning interface, and automated dispatch engines (cellular SMS via Twilio API, SMTP email, acoustic sirens, and physical optocoupled dry-contact industrial relays).

- **Power & Backup System Resilience:** Validated a 24.5 hour continuous wearable battery lifespan (1100 mAh LiPo, Factor of Safety = 2.04 for 12 h shifts) and demonstrated 9 hours 52 minutes of autonomous server survivability during total building power blackout.

Commercial Readiness & Strategic Roadmap

The prototype hardware baseline has been validated at a total unit procurement cost of \$966.10 AUD. Transitioning to the commercialisation phase (structured over a 60-day industrial engagement), the platform will consolidate all wearable circuitry onto a custom monolithic 4-layer SMT printed circuit board (reducing per-unit wearable BOM to < \$32.00 AUD at scale), enclosed within an IP67-rated drop-resistant polycarbonate housing. The roadmap includes 30-day live customer pilot deployments (e.g., Goldstream RV manufacturing plant) and strategic scoping for autonomous drone-assisted outdoor emergency verification.

Contents

List of Figures

List of Tables

1. Introduction

1.1 Problem Statement & Motivation

In modern industrial, manufacturing, healthcare, and logistics facilities, ensuring the safety of personnel is a vital operational and legal priority. Organizations are mandated under statutory Work Health and Safety (WHS) frameworks—such as the Safe Work Australia Model Code of Practice for Remote or Isolated Work [1] and international standard ISO 45001 [2]—to provide effective communication systems and rapid emergency response mechanisms for workers. This duty of care is particularly acute for “lone workers” and isolated staff members who operate without direct or close supervision, often in acoustically shielded or physically compartmentalized industrial environments.

A representative operational case study is provided by manufacturing facilities such as Goldstream RV (Pakenham, Victoria). In this single-level, U-shaped industrial manufacturing plant constructed of heavy steel sheet over structural framework, several critical hazard scenarios routinely manifest:

1. **Isolated Personnel Incapacitation:** Assembly technicians frequently operate independently inside enclosed recreational vehicles (RVs) for hours at a time. In the event of an acute medical collapse, fall, or traumatic injury, an incapacitated worker may remain undetected until end-of-day site closure.
2. **Evacuation Unaccountability:** During facility-wide emergencies (such as fires or chemical spills), independent personnel working inside acoustically insulated cabins or remote workshops may fail to hear building sirens. Safety wardens are forced to conduct high-risk physical sweeps across complex layouts without knowing real-time worker locations.
3. **Irregular Shifts & Contractor Tracking:** Staff working overtime or irregular weekend shifts may be the sole occupants of large facility sectors. Furthermore, visiting contractors and transient maintenance personnel often operate outside standard automated payroll/bundy-clock systems, leading to hazardous discrepancies in real-time site occupancy records.

Existing commercial lone-worker safety solutions predominantly rely on standalone cellular wearables tethered to proprietary, recurring Software-as-a-Service (SaaS) subscription models (e.g., Blackline Safety, SoloProtect) [3]. These systems present severe drawbacks for industrial deployment:

- **Prohibitive Operating Expenditures:** Cumulative annual per-device subscription fees create substantial financial friction for small-to-medium enterprises (SMEs) operating on fixed capital expenditure (CAPEX) cycles.
- **Cellular Dead Zones & Penetration Loss:** Industrial facilities characterized by reinforced concrete, metallic cladding, and internal machinery suffer from significant RF attenuation, rendering public cellular networks unreliable indoors.

- **Lack of Integrated Physical Infrastructure:** Commercial wearables typically operate as isolated signaling beacons without direct hardware hooks into local facility infrastructure (e.g., automated site evacuation sirens, physical dry-contact relays, or localized zoning maps).

Consequently, there is an urgent engineering imperative for an on-premises, dual-redundant, subscription-free Emergency Response System (ERS) that unifies wearable biometric/kinematic sensing, low-power long-range wireless telemetry, real-time indoor positioning, and automated multi-channel facility dispatch.

1.2 Project Background & Developmental Trajectory

The development of the Micromax Emergency Response System has advanced across three sequential phases, evolving from an exploratory academic capstone project into a commercially viable industrial safety platform:

1. **Phase 1 — Academic Proof-of-Concept (ECTE351):** Initiated as an engineering capstone design project at the University of Wollongong in collaboration with Micromax Technology (Team 11), this phase established baseline feasibility. It demonstrated wearable fall detection via a 6-axis inertial measurement unit (IMU), basic cellular SMS notifications, and a conceptual HTML monitoring dashboard.
2. **Phase 2 — Functional Prototype Engineering (ECTE399 R&D):** Conducted as an extensive Research and Development internship at Micromax Technology under engineering mentorship, this phase transitioned the system from a breadboard concept to a deployable, multi-node functional prototype. Major engineering milestones achieved include:
 - Implementation of a dual-redundant wireless communication architecture coupling primary LoRaWAN (AU915) long-range telemetry with automated fallback to local IEEE 802.11 Wi-Fi.
 - Deployment of an indoor Real-Time Location System (RTLS) utilizing Bluetooth Low Energy (BLE) gateway infrastructure, RSSI path-loss filtering (Kalman filtering), and symbolic Room Zone classification.
 - Development of multi-modal client emergency logic (personal panic vs. mass facility emergency) and an MPU6050 fall detection algorithm with an audible false-alarm cancellation grace window.
 - Engineering of a centralized Raspberry Pi 5 edge server hosting an SQLite relational database, multi-channel dispatch engine (SMS, Email, audio sirens, dry-contact hardware relays), and a web-based floor plan zoning interface.
 - Power budget characterization and Uninterruptible Power Supply (UPS) power-sequencing integration ensuring > 9.8 hours of autonomous server survivability and full-shift wearable endurance.

3. **Phase 3 — Commercialisation & Product Refinement:** The ongoing industrialisation phase focuses on transforming the lab-tested prototype into a pilot-ready, ruggedized commercial product. This encompasses custom 4-layer PCB design (integrating RP2040 processing, LoRa transceivers, and BLE beaconing into an ergonomic wearable form factor), IP-rated mechanical enclosure tooling, manufacturing handover documentation, and strategic scoping for autonomous drone-assisted facility inspection.

1.3 Project Scope & Core Engineering Objectives

This technical report documents the complete systems engineering design, empirical validation, and commercialisation pathway of the Micromax Emergency Response System. The primary engineering objectives addressed in this work are:

1. **Objective 1: Multi-Modal Active and Passive Emergency Detection**

Design, calibrate, and validate embedded firmware on the client wearable capable of differentiating between deliberate Personal Emergencies (press-and-hold), facility-wide Mass Emergencies (rapid multi-click burst), and passive traumatic falls (IMU acceleration/gyroscope thresholding with false-alarm cancellation).

2. **Objective 2: Fail-Safe Dual-Redundant Wireless Telemetry**

Formulate and implement a communication protocol stack prioritizing low-power, sub-gigahertz LoRaWAN transmission for reliable penetration through industrial steel structures, coupled with autonomous failover to local Wi-Fi upon packet acknowledgement failure.

3. **Objective 3: Robust Real-Time Indoor Localization**

Evaluate and deploy a multi-gateway BLE RSSI tracking network. Analyze trilateration, centroid estimation, Kalman filtering, and fuzzy logic techniques to achieve high-confidence room-level zoning on interactive facility floor plans.

4. **Objective 4: Centralized Edge Telemetry, Automation & Audit Logging**

Construct an edge computing server environment that translates compact binary radio payloads, manages dynamic user/device allocations, coordinates multi-channel dispatch (cellular SMS, SMTP email, acoustic alarms, industrial relay outputs), and maintains tamper-evident audit logs.

5. **Objective 5: Commercial Viability, DFM & Strategic Scalability**

Establish a comprehensive Bill of Materials (BOM), analyze system power budgets under real-world discharge profiles, identify hardware failure modes, and delineate a concrete manufacturing and commercialisation roadmap.

1.4 Report Organization

The remainder of this report is organized as follows:

- **Chapter ?? (Background & Literature Review):** Details the mathematical foundations of kinematic fall detection, RF path-loss propagation, Kalman filter state estimation, fuzzy inference systems, LoRaWAN network dynamics, and relevant occupational health and safety standards.
- **Chapter ?? (System Architecture & Methodology):** Presents the comprehensive multi-tier hardware, embedded firmware, communication protocol, and server software architectures, including circuit-level design and web dashboard workflows.
- **Chapter ?? (Results & Performance Analysis):** Provides empirical validation data, including fall detection calibration matrices, BLE gateway attenuation profiles, indoor localization accuracy benchmarks, battery discharge kinetics, server UPS reliability metrics, and the final Bill of Materials.
- **Chapter ?? (Discussion):** Evaluates architectural trade-offs, industrial deployment challenges within metallic environments, comparison against subscription models, and outlines the commercialisation roadmap.
- **Chapter ?? (Conclusion & Future Work):** Summarizes major engineering achievements, verifies deliverable compliance, and outlines future technical directions including edge-AI and drone integration.
- **Appendices:** Contains critical firmware source code listings, packet parsing scripts, database schemas, and complete procurement datasets.

2. Theoretical Background & Literature Review

2.1 Theoretical Foundations

2.1.1 Kinematic Principles of Fall Detection

Automated fall detection relies on tri-axial inertial measurement unit (IMU) sensing to detect the distinct physical phases of human traumatic falls: initial freefall/weightlessness, high-g kinetic impact upon ground contact, and subsequent post-fall immobility (prostration) [4].

Let the instantaneous linear accelerations along the orthogonal axes of a body-worn sensor be denoted by $a_x(t)$, $a_y(t)$, and $a_z(t)$ in units of gravitational acceleration ($g \approx 9.81 \text{ m/s}^2$). The total Acceleration Vector Magnitude (AVM), denoted as $A_{\text{total}}(t)$, is expressed by:

$$A_{\text{total}}(t) = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2} \quad (2.1)$$

Similarly, for the tri-axial angular rates $\omega_x(t)$, $\omega_y(t)$, and $\omega_z(t)$ measured by a gyroscope in degrees per second ($^\circ/\text{s}$), the total Angular Velocity Magnitude $\Omega_{\text{total}}(t)$ is defined as:

$$\Omega_{\text{total}}(t) = \sqrt{\omega_x(t)^2 + \omega_y(t)^2 + \omega_z(t)^2} \quad (2.2)$$

A traumatic fall sequence manifests three deterministic kinematic signatures:

1. **Freefall Phase:** Prior to ground collision, the acceleration magnitude experiences a transient valley below gravity:

$$A_{\text{total}}(t) < \text{Th}_{\text{freefall}} \quad (\text{typically } \approx 0.5g) \quad (2.3)$$

2. **Impact Shock Phase:** Collision with the floor results in a rapid mechanical deceleration producing a sharp acceleration spike:

$$A_{\text{total}}(t) \geq \text{Th}_{\text{impact}} \quad (\text{typically } \geq 2.5g - 3.0g) \quad (2.4)$$

3. **Postural Inactivity & Prostration:** Following impact, if the worker is incapacitated, the device settles into a stationary horizontal orientation. The body tilt angle $\theta_{\text{tilt}}(t)$ relative to the vertical gravity vector is calculated from the low-pass filtered static acceleration components:

$$\theta_{\text{tilt}}(t) = \arccos\left(\frac{a_z(t)}{A_{\text{total}}(t)}\right) \quad (2.5)$$

A prolonged post-impact state where $|\theta_{\text{tilt}}| > \text{Th}_{\text{prostration}}$ and dynamic variance $\sigma_A^2 < \epsilon$ for a duration $\Delta t \geq 1.5 \text{ s}$ confirms a collapse event.

2.1.2 Radio Frequency Propagation & Path Loss Modeling

In wireless indoor localization systems, the Received Signal Strength Indicator (RSSI) measured in decibels relative to one milliwatt (dBm) is a non-linear function of Euclidean distance between the transmitter and receiver [5]. In enclosed industrial spaces, signal attenuation is governed by the Log-Distance Path Loss Model:

$$\text{RSSI}(d) = \text{RSSI}(d_0) - 10 n \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma \quad (2.6)$$

where:

- d is the Euclidean distance between transmitter and receiver (meters),
- d_0 is the reference distance (standardized at $d_0 = 1$ m),
- $\text{RSSI}(d_0)$ (or A) is the measured signal strength at 1 m (typically -50 dBm to -65 dBm),
- n is the Path Loss Exponent (PLE), representing environmental attenuation ($n = 2.0$ for free-space, $n = 2.5 - 3.5$ for office spaces, and $n = 3.5 - 5.0$ in cluttered industrial environments with metallic obstacles),
- $X_\sigma \sim \mathcal{N}(0, \sigma^2)$ is a zero-mean Gaussian random variable capturing shadow fading and multipath reflections.

Inverting Equation (??) yields the estimated distance $\hat{d}$ from a measured RSSI value:

$$\hat{d} = d_0 \cdot 10^{\frac{\text{RSSI}(d_0) - \text{RSSI}}{10 n}} \quad (2.7)$$

2.1.3 Geometric Localization: Trilateration & Weighted Centroid

Given a network of $M \geq 3$ fixed BLE gateways located at known 2D Cartesian coordinates (x_i, y_i) with corresponding estimated distances $\hat{d}_i$, the target node position (x, y) satisfies the non-linear system of intersecting circles [6]:

$$(x - x_i)^2 + (y - y_i)^2 = \hat{d}_i^2, \quad \forall i \in \{1, 2, \dots, M\} \quad (2.8)$$

Linearizing Equation (??) by subtracting the M -th circle equation from the preceding $M - 1$ equations yields the linear matrix equation $\mathbf{A}\mathbf{x} = \mathbf{b}$:

$$2 \begin{bmatrix} x_1 - x_M & y_1 - y_M \\ x_2 - x_M & y_2 - y_M \\ \vdots & \vdots \\ x_{M-1} - x_M & y_{M-1} - y_M \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (x_1^2 - x_M^2) + (y_1^2 - y_M^2) + (\hat{d}_M^2 - \hat{d}_1^2) \\ (x_2^2 - x_M^2) + (y_2^2 - y_M^2) + (\hat{d}_M^2 - \hat{d}_2^2) \\ \vdots \\ (x_{M-1}^2 - x_M^2) + (y_{M-1}^2 - y_M^2) + (\hat{d}_M^2 - \hat{d}_{M-1}^2) \end{bmatrix} \quad (2.9)$$

The optimal least-squares position estimate $\hat{\mathbf{x}} = [\hat{x}, \hat{y}]^T$ is solved via Moore-Penrose pseudo-inversion:

$$\hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} \quad (2.10)$$

In scenarios where geometric trilateration becomes ill-conditioned due to co-linear gateway placement or extreme multipath fading, Weighted Centroid Localization (WCL) provides a robust fallback:

$$\hat{x} = \frac{\sum_{i=1}^M w_i x_i}{\sum_{i=1}^M w_i}, \quad \hat{y} = \frac{\sum_{i=1}^M w_i y_i}{\sum_{i=1}^M w_i}, \quad \text{where } w_i = \frac{1}{\hat{d}_i^p} \text{ or } w_i = 10^{\frac{\text{RSSI}_i}{10}} \quad (2.11)$$

2.1.4 State Estimation via Discrete Kalman Filtering

To suppress high-frequency fluctuations and multipath noise in dynamic RSSI measurements, a discrete one-dimensional Kalman Filter is applied to raw signal streams prior to distance transformation [7]. The state space is formulated as:

$$x_k = x_{k-1} + w_k, \quad w_k \sim \mathcal{N}(0, Q) \quad (2.12)$$

$$z_k = x_k + v_k, \quad v_k \sim \mathcal{N}(0, R) \quad (2.13)$$

where x_k represents the true underlying RSSI state at time step k , z_k is the raw measured RSSI, Q is the process noise covariance, and R is the measurement noise covariance.

The recursive two-step Kalman estimation algorithm executes as follows:

1. Time Update (Predict):

$$\hat{x}_k^- = \hat{x}_{k-1} \quad (2.14)$$

$$P_k^- = P_{k-1} + Q \quad (2.15)$$

2. Measurement Update (Correct):

$$K_k = \frac{P_k^-}{P_k^- + R} \quad (2.16)$$

$$\hat{x}_k = \hat{x}_k^- + K_k (z_k - \hat{x}_k^-) \quad (2.17)$$

$$P_k = (1 - K_k)P_k^- \quad (2.18)$$

Here, $K_k \in [0, 1]$ is the dynamic Kalman gain, which balances responsiveness against high-frequency noise rejection.

2.1.5 Fuzzy Logic Inference for Room-Level Classification

Due to the non-linearities and spatial anomalies inherent to indoor RF environments, converting filtered RSSI values into absolute coordinate pairs (x, y) can produce spatial errors exceeding room dimensions. To address this, a Mamdani Fuzzy Inference System (FIS) is integrated to map signal vector profiles directly into discrete room probabilities [8].

The fuzzy inference engine processes three primary linguistic inputs for each gateway:

- **Signal Strength (μ_{RSSI}):** Partitioned into fuzzy sets {Weak, Medium, Strong, Near-Field}.
- **Signal Variance (σ_{RSSI}):** Evaluates environmental stability as {Stable, Fluctuating}.

- **Inter-Gateway RSSI Delta ($\Delta\text{RSSI}_{ij} = \text{RSSI}_i - \text{RSSI}_j$):** Evaluates relative spatial proximity.

A typical rule base executes as:

$\mathcal{R}_k$: **IF** RSSI_1 is Strong AND RSSI_2 is Weak AND $\Delta\text{RSSI}_{12} > 15$ dB **THEN** Zone is Office 1

Defuzzification using the Center of Gravity (CoG) method yields a crisp confidence index for each mapped Room Zone, ensuring robust spatial classification even when line-of-sight is obstructed by structural walls.

2.1.6 Wireless Protocol Architecture: LoRaWAN vs. Wi-Fi

The dual-redundant communication backbone of the ERS balances the distinct operational advantages of Chirp Spread Spectrum (CSS) LoRaWAN and Orthogonal Frequency-Division Multiplexing (OFDM) Wi-Fi (IEEE 802.11 b/g/n) [9]:

- **LoRaWAN (Sub-GHz, AU915 Band):** Operates with a wide link budget (> 150 dB) and robust obstacle penetration through metallic machinery and structural steel. It utilizes an uncoordinated ALOHA channel access model with spreading factors $SF \in \{7, 8, \dots, 12\}$. For an N -byte payload, the total packet airtime T_{packet} is:

$$T_{\text{packet}} = T_{\text{preamble}} + T_{\text{payload}} = (N_{\text{preamble}} + 4.25) T_{\text{sym}} + N_{\text{payload_sym}} T_{\text{sym}} \quad (2.19)$$

where the symbol duration $T_{\text{sym}} = \frac{2^{SF}}{\text{BW}}$. To minimize collision probability and maximize battery life, payloads must be compressed into minimal binary representations.

- **Wi-Fi (2.4 GHz IEEE 802.11):** Provides high-throughput (> 10 Mbps), low-latency (< 50 ms) IP-level connectivity with standard Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA). However, 2.4 GHz carrier frequencies suffer from severe attenuation through heavy industrial cladding and require higher sustained transceiver current (80 – 120 mA active vs. < 25 mA for LoRa).

2.2 Industrial Standards & Competitive Landscape

2.2.1 Occupational Health & Safety Compliance Framework

Under the Safe Work Australia Model Code of Practice [1], employers are legally mandated to implement systems of work that:

1. Provide effective communication systems with workers who work alone or in remote areas.
2. Establish scheduled check-in and automated fail-safe distress escalation protocols.
3. Maintain rapid emergency dispatch capabilities with precise location tracking to ensure medical assistance within critical response windows.

Furthermore, compliance with ISO 45001:2018 Clause 8.2 (Emergency Preparedness and Response) necessitates verified periodic testing, incident tracking, and documented audit trails [2].

2.2.2 Competitive Solution Benchmarking

Table ?? benchmarks the Micromax Emergency Response System against leading commercial enterprise lone-worker solutions [3].

Table 2.1: Commercial Lone-Worker Safety System Comparison

Feature / Parameter	Micromax ERS	Blackline Safety G7c	SoloProtect ID Touch	Wearin' Connected
Commercial Model	One-off CAPEX purchase	SaaS subscription (\$30–60/mo/unit)	SaaS subscription (\$25–50/mo/unit)	Enterprise bundle + SaaS
Primary Wireless	LoRaWAN (AU915, 8-Ch)	Cellular (4G LTE-M / 2G)	Cellular (4G / VoLTE)	Cellular + Mesh BLE
Backup Wireless	IEEE 802.11 b/g/n Wi-Fi	Satellite (add-on module)	None	Local BLE gateway
Indoor Positioning	BLE Gateways + Room Zones	Location Beacons (Cellular)	BLE / Wi-Fi Sniffing	Proprietary AoA Beacons
Fall Detection	6-DOF IMU + 10s grace	3-Axis Accelerometer	Accelerometer (Man-Down)	IMU + Biometrics
Facility Siren/Relay	Direct dry-contact relay	API webhook / external	Cloud API only	Cloud API only
Data Ownership	On-premises SQLite / Edge	Cloud hosted (Vendor)	Cloud hosted (Vendor)	Cloud / Hybrid
Drill / Audit Mode	Integrated web drill mode	Manual log flagging	Portal export	Specialized software

As demonstrated in Table ??, the Micromax ERS eliminates recurring subscription overheads while providing localized hardware redundancy, on-premises compliance auditing, and native facility hardware actuation.

3. System Architecture & Engineering Methodology

3.1 System Architecture Overview

The Micromax Emergency Response System is architected as a distributed, four-tier Internet of Things (IoT) edge-processing platform designed for mission-critical workplace safety. As illustrated in the block topology, the system comprises:

1. **Wearable Client Nodes:** Microcontroller-driven wearable safety tags equipped with multi-modal emergency triggering, 6-DOF inertial fall sensing, dual-band wireless telemetry (LoRaWAN / Wi-Fi), and periodic BLE location beaconing.
2. **Ingress Gateway Infrastructure:** A dual-path gateway array consisting of an 8-channel SX1302/SX1303 LoRaWAN concentrator for telemetry packet reception and a network of distributed BLE-to-Wi-Fi/Ethernet gateways (Minew G1 / Mokosmart MKGW3) for RSSI harvesting.
3. **Centralized Edge Processing Server:** A dedicated Raspberry Pi 5 unit hosting a multithreaded Python backend, local UDP packet forwarder, MQTT message broker, SQLite relational database, and real-time positioning algorithms.
4. **Automated Notification & Dispatch Layer:** A multi-channel alert engine interfacing with external cellular SMS gateways (Twilio API), SMTP mail relays, local acoustic annunciators, and physical dry-contact relays for facility evacuation sirens.

3.2 Hardware Design & Component Selection

3.2.1 Wearable Client Device Hardware

The client node is designed to balance low power consumption, processing capability, sensor fidelity, and compact physical dimensions. The hardware architecture incorporates the following integrated modules:

- **Core Microcontroller (Raspberry Pi Pico 2WH):** Powered by the RP2040 dual-core ARM Cortex-M0+ processor operating at 133 MHz with 264 KB of on-chip SRAM and 2 MB of onboard QSPI flash memory [10]. The onboard CYW43439 wireless chip provides secondary 2.4 GHz IEEE 802.11 b/g/n Wi-Fi connectivity.
- **LoRaWAN Transceiver Module (Waveshare Pico LoRa HAT):** Integrates a Semtech SX1262 sub-GHz RF transceiver tuned for the Australian AU915-928 MHz Industrial, Scientific, and Medical (ISM) band. It interfaces with the RP2040 via high-speed SPI with dedicated interrupt lines (DIO1) and an external 915 MHz RP-SMA dipole antenna.

- **Dedicated BLE Beacons Module (Seeed Studio XIAO nRF52840):** Featuring a Nordic nRF52840 32-bit ARM Cortex-M4F processor with Bluetooth 5.0 Low Energy capability. The module continuously broadcasts non-connectable advertising beacons (iBeacon standard, 100 ms interval, 0 dBm Tx power) for gateway RSSI capture without placing computational overhead on the primary RP2040 processor.
- **Inertial Measurement Unit (InvenSense MPU6050):** A 6-DOF sensor combining a 3-axis MEMS accelerometer ($\pm 8g$ range, 4096 LSB/ g) and a 3-axis MEMS gyroscope ($\pm 1000^\circ/s$ range, 32.8 LSB/ $(^\circ/s)$) connected via I2C at 400 kHz (Fast Mode) on GPIO4 (SDA) and GPIO5 (SCL) [11].
- **Power Management & Battery Subsystem:** A single-cell rechargeable Lithium-Polymer (LiPo) battery rated at 3.7 V nominal, 1100 mAh capacity (4.07 Wh), coupled with a WaveShare Pico UPS Module (Type B) incorporating a Texas Instruments step-up boost converter (5.0 V rail) and an I2C fuel gauge for dynamic cell voltage monitoring.
- **User Interface & Physical Actuation:** A high-tactile snap-action push button with mechanical debouncing, a high-intensity dual-color status LED (green for operational status, pulsing red for emergency trigger), and an active piezoelectric acoustic buzzer (85 dB @ 10 cm) for pre-alert audible warnings.

3.2.2 Central Edge Server & Gateway Layer

- **Central Edge Server (Raspberry Pi 5):** Features a Broadcom BCM2712 quad-core ARM Cortex-A76 processor @ 2.4 GHz, 4 GB LPDDR4X SDRAM, Gigabit Ethernet, and the custom RP1 I/O controller handling low-level peripherals [12].
- **LoRaWAN Concentrator HAT (WaveShare SX1302/SX1303 915M):** An industrial 8-channel gateway baseband processor capable of simultaneous multi-SF packet demodulation across 8 frequency channels in the AU915 band (FSB 2, 916.8 – 918.2 MHz), providing a receiver sensitivity down to -141 dBm [13].
- **Distributed BLE Gateways:** Minew G1 BLE-to-Wi-Fi/Ethernet gateways and Mokosmart MKGW3 PoE BLE gateways deployed in fixed ceiling/wall locations. The gateways continuously scan 2.4 GHz BLE advertising channels (37, 38, 39), extract transmitter MAC addresses, parse payload UUIDs, and forward structured JSON payloads containing raw RSSI values to the server’s central MQTT broker over local LAN.

Table ?? summarizes the operating parameters of the ERS infrastructure.

3.3 Firmware Architecture & Embedded Logic

3.3.1 Multi-Modal Button Triggering State Machine

To eliminate false alarms while enabling intuitive single-button operation, the client firmware implements a deterministic state machine distinguishing between two primary user-initiated alert classes:

Table 3.1: Emergency Response System Technical Specifications

Subsystem / Parameter	Technical Specification / Description	Nominal Value
Client Microcontroller	Raspberry Pi Pico 2WH (RP2040, Dual Cortex-M0+)	133 MHz, 264 KB SRAM
Primary Wireless Radio	Semtech SX1262 LoRa Transceiver (SPI)	915 MHz AU915 Band
Secondary Wireless Radio	Infineon CYW43439 Wi-Fi (IEEE 802.11 b/g/n)	2.4 GHz, TCP/UDP Sockets
Location Beacon Radio	Nordic nRF52840 BLE 5.0 (Advertising Mode)	2.4 GHz, 100 ms Interval
Inertial Sensor	InvenSense MPU6050 6-DOF IMU (I2C)	$\pm 8g$, $\pm 1000^\circ/\text{s}$, 50 Hz
Client Battery Cell	Rechargeable LiPo 1S Pouch Cell	3.7 V, 1100 mAh (4.07 Wh)
Client Power Budget	Active Beacons + Idle Listen Current	42 mA Avg (> 24 h runtime)
Server Platform	Raspberry Pi 5 Model B (Quad Cortex-A76)	2.4 GHz, 4 GB RAM
LoRaWAN Concentrator	Semtech SX1302/SX1303 8-Channel HAT (SPI)	$8 \times$ Uplink, -141 dBm Sens.
Gateway Power Supply	Power over Ethernet (PoE IEEE 802.3af) / 5V DC	5 V/1 A
Server UPS Battery	Waveshare Pi 5 UPS Hat (2 $\times$ 18650 Li-ion)	7.4 V, 5000 mAh (> 9.8 h backup)
Operating Temperature	Industrial Ambient Temperature Range	-10°C to $+55^\circ\text{C}$

1. **Personal Emergency (Individual Distress):** Activated by a continuous, sustained button press for $T_{\text{hold}} \geq 2.0$ seconds. This triggers a localized distress alert routed specifically to assigned safety supervisors without sounding facility-wide sirens.
2. **Mass Facility Emergency (Building Evacuation):** Activated by a rapid sequence of $N_{\text{press}} = 5$ consecutive button presses completed within a total window of $T_{\text{burst}} \leq 3.0$ seconds. This triggers a critical building evacuation workflow, actuating the central siren and notifying all site personnel.

3.3.2 Hardware Conflict Identification & Resolution (GPIO15 Debug)

During initial prototype integration, a critical hardware bug was observed: while Personal Emergency alerts transmitted reliably over LoRaWAN, Mass Emergency triggers consistently failed over LoRa and were prematurely forced into Wi-Fi failover.

Engineering investigation revealed that GPIO15 on the RP2040 had been assigned both as the external push-button interrupt input and the hardware active-low reset pin (**NRESET**) of the SX1262 LoRa module. When a user executed 5 rapid button presses, GPIO15 was repeatedly pulled low and released, subjecting the SX1262 to consecutive hardware resets during SPI initialization. This corrupted the internal radio state machine and caused LoRa transmission timeouts. The issue was resolved by remapping the LoRa **NRESET** line to an isolated GPIO (GPIO13) and adding a 100 nF RC hardware debounce circuit to the button input on GPIO15.

3.3.3 Kinematic Fall Detection Algorithm with Cancellation Grace Window

The MPU6050 driver continuously samples acceleration and angular rate registers into a circular buffer at a frequency of 50 Hz ($\Delta t = 20$ ms). The algorithm operates across four sequential states:

1. **State 0 (Continuous Monitoring):** Calculates $A_{\text{total}}(t)$ via Equation (??). If $A_{\text{total}}(t) < 0.5g$ (freefall detected) followed within 300 ms by $A_{\text{total}}(t) \geq 2.8g$ (impact shock), the state machine advances to State 1.
2. **State 1 (Orientation & Inactivity Verification):** The microcontroller monitors sensor variance for 1.5 seconds. If post-impact dynamic variance $\sigma_A^2 < 0.05g^2$ and tilt angle $|\theta_{\text{tilt}}| > 60^\circ$ (prostration confirmed), a fall event is registered, transitioning to State 2.
3. **State 2 (Cancellation Grace Window):** To prevent false alarms arising from dropped devices or heavy tool placement, the client initiates a 10.0 second countdown. During this window, the local buzzer emits an intermittent high-pitch warning beep (2 Hz) and the status LED flashes red. If the user presses the button during this 10-second period, the alert is cancelled, logged locally as a false positive, and normal monitoring resumes.
4. **State 3 (Distress Dispatch):** If the countdown expires without user cancellation, the firmware compiles a high-priority distress packet and passes it to the transmission subsystem.

3.3.4 Compact 4-Byte LoRaWAN Binary Payload Design

To conform to LoRaWAN regional duty cycle regulations (maximum 1% airtime in ISM bands) and minimize packet collision probability, all telemetry is serialized into an ultra-compact 4-byte binary structure, detailed in Table ??.

Table 3.2: Compact 4-Byte LoRaWAN Telemetry Payload Structure

Byte Index	Bit Field	Data Type	Description & Bit Mapping
Byte 0	Bits 7–0	uint8_t	Client Device ID: Unique numerical identifier ($0 \times 01 - 0 \times FF$, supporting 255 local nodes).
Byte 1	Bits 7–4	uint4_t	Event Type Code: 0×1 : Personal Emergency, 0×2 : Mass Evacuation, 0×3 : Traumatic Fall, 0×4 : Low Battery Warning, 0×0 : Periodic Heartbeat / Keepalive.
	Bits 3–0	uint4_t	Battery Charge State: Scaled 0 – 15 representing 0% – 100% capacity (6.67% quantization step).
Byte 2	Bits 7–0	uint8_t	Sequence Counter (High Byte): Rolling transaction counter (0–65535) for packet deduplication.
Byte 3	Bits 7–0	uint8_t	Sequence Counter (Low Byte) & Status Flags: Includes acknowledge bit and failover status flags.

3.3.5 Autonomous LoRaWAN-to-Wi-Fi Failover State Machine

The client communication subsystem implements a dual-link failover engine (encapsulated in `ClientDeviceV2.py`):

1. When an emergency event triggers, the client formats the 4-byte payload and transmits it over LoRaWAN (AU915, SF7, 125 kHz bandwidth).
2. The client opens an RX1 receive window (1.0s post-transmission) awaiting a server acknowledgement (ACK) frame.
3. If no ACK is received, the client executes a second LoRa retry ($N_{\text{retry}} = 2$).
4. If the secondary LoRa transmission times out (indicating RF occlusion, concentrator outage, or deep steel attenuation), the microcontroller immediately activates the onboard CYW43439 Wi-Fi radio, connects to the local emergency SSID, and transmits the alert payload via a raw TCP/UDP socket directly to the server IP on port 5005.

3.4 Server Architecture, Database Schema & Dispatch Pipeline

3.4.1 Edge Server Multithreaded Software Architecture

The Raspberry Pi 5 runs a custom Linux service suite orchestrating concurrent network protocols:

- **LoRa Packet Forwarder (`lora_pkt_fwd`):** Interfaces with the SX1302 concentrator over SPI, receiving raw radio packets and forwarding them via local UDP socket to `local_lora_udp_server.py`.
- **MQTT Broker (Eclipse Mosquitto):** Listening on `192.168.115.49:1883`, receiving JSON RSSI streams from the distributed Minew G1 and Mokosmart gateways.
- **Core Server Engine (`server.py`):** Implements multi-threading to parse incoming LoRa and Wi-Fi packets, query the database, calculate node positions, and trigger notification workflows.
- **Web Application (Flask / WSGI):** Serves the responsive management dashboard on port 5000.

3.4.2 SQLite Relational Database Schema

The persistence layer utilizes SQLite3 to ensure zero-maintenance, zero-configuration local data logging. The relational structure is defined across four core tables:

1. **staff:** Stores `staff_id` (PRIMARY KEY), `full_name`, `email`, `phone_number`, `department`, `special_medical_notes`, and `is_active`.
2. **devices:** Stores `device_id` (PRIMARY KEY), `lora_device_id`, `ble_mac_address`, `assigned_staff_id` (FOREIGN KEY), `battery_percentage`, and `last_heartbeat`.
3. **room_zones:** Stores `zone_id` (PRIMARY KEY), `zone_name`, `x_min`, `y_min`, `x_max`, `y_max` (defining bounding polygon coordinates on the floor plan), and `floor_level`.
4. **incident_logs:** Stores `log_id` (PRIMARY KEY), `timestamp`, `device_id`, `event_type`, `raw_rssi_data`, `estimated_x`, `estimated_y`, `resolved_room_name`, `dispatch_status`, and `is_drill`.

3.4.3 Multi-Channel Notification & Actuation Pipeline

Upon receipt of a validated emergency packet, `server.py` instantiates parallel asynchronous dispatch workers:

- **Cellular SMS Dispatch (Twilio API):** Transmits formatted SMS alerts to configured emergency response numbers containing worker name, event classification, timestamp, and resolved location (e.g., “*EMERGENCY: Fall Detected for John Doe at 14:22:05. Location: Office 1 (Boardroom Zone). Immediate assistance required.*”)
- **SMTP Email Dispatch:** Generates MIME-formatted HTML emails containing detailed incident summaries, worker medical notes, and floor plan coordinate links.
- **Hardware Siren & Relay Output:** Asserts GPIO26 on the Raspberry Pi 5, closing an optocoupled industrial dry-contact relay (250 V AC/10 A) connected to building evacuation sirens and strobes. A dedicated “*Acknowledge / Silence Siren*” button on the web GUI de-asserts the relay once safety wardens verify site control.

3.4.4 Dynamic Web Management GUI & Interactive Floor Plan Mapper

The web interface provides real-time facility visibility split across four dedicated modules:

1. **Live Dashboard:** Displays active worker cards, battery telemetry, real-time connectivity status (Green = Normal, Red = Alarm, Grey = Offline), and dynamic SVG floor plan markers.
2. **Interactive Room Zone Mapper:** Allows administrators to upload high-resolution facility floor plans (PNG/JPEG), calibrate pixel-to-meter scaling by setting known anchor dimensions, place virtual gateway coordinates, and define Room Zones by clicking two bounding corners on the canvas.
3. **Staff & Device Allocation:** Facilitates assignment of wearable hardware to permanent staff or temporary visitors with automatic data purge timers upon visitor check-out.
4. **Audit Logging & Evacuation Drill Mode:** Provides CSV log exporting for compliance audits and an evacuation drill mode that executes full dispatch workflows while tagging database entries as `is_drill = 1` to prevent false regulatory reporting.

3.5 Experimental Testbed & Calibration Methodology

Empirical evaluation was conducted within the Micromax Technology facility (5 Orangegrove Ave, Unanderra, NSW). The testing methodology encompassed:

- **Fall Detection Calibration:** Controlled drop trials onto a high-density safety mattress across 5 distinct impact angles (forward collapse, backward fall, lateral left/right, and vertical collapse) compared against 5 Activities of Daily Living (ADL: brisk walking, sitting rapidly, bending, stair descent, and prayer prostration).
- **BLE Gateway Range & Attenuation Profiling:** Stepped distance measurements (0 m to 7 m) across open-air, drywall, and structural steel partitions comparing the Minew G1, Mokosmart MKGW3, and Mokosmart LW003-B.
- **Power & UPS Endurance Testing:** Automated voltage and current discharge profiling under active 100 ms BLE advertising and continuous server workloads down to hardware cutoff.

4. Results & Performance Analysis

4.1 Fall Detection Calibration & Kinematic Validation

To evaluate the classification fidelity of the MPU6050 fall detection algorithm and calibrate optimal detection thresholds, empirical trials were conducted comparing traumatic fall scenarios against common Activities of Daily Living (ADL).

Table ?? presents the recorded peak tri-axial accelerations, total vector magnitude (A_{total}), angular velocities (G_x, G_y, G_z), and Euler orientation angles (Pitch, Roll, Yaw) across controlled physical trials.

Table 4.1: Kinematic Sensor Calibration Dataset for Fall Events vs. Daily Living Activities

Activity / Trial	A_x (g)	A_y (g)	A_z (g)	A_{total} (g)	G_x	G_y	G_z	Pitch	Roll	Yaw
Forward Fall	2.45	0.85	1.90	3.21	4.12	3.85	1.20	1.42	0.31	0.55
Backward Fall	1.20	0.65	2.85	3.16	3.95	4.50	0.95	1.51	0.22	0.48
Lateral Fall (Left)	0.95	2.60	1.50	3.15	2.80	1.10	4.20	0.35	1.48	0.62
Lateral Fall (Right)	0.88	2.75	1.42	3.21	2.95	1.25	4.35	0.32	1.52	0.59
Vertical Collapse	0.62	0.55	2.95	3.06	1.85	2.10	1.15	1.38	0.40	0.30
Prayer Prostration	0.60	0.20	0.40	0.75	0.51	0.56	0.50	0.61	0.24	0.35
Rapid Sitting	0.35	0.25	1.65	1.70	1.10	0.85	0.40	0.25	0.15	0.20
Brisk Walking	0.45	0.30	1.25	1.36	0.95	1.20	0.80	0.18	0.12	0.45
Stair Descent	0.55	0.40	1.80	1.92	1.35	1.10	0.65	0.22	0.18	0.30
Bending to Floor	0.70	0.30	0.65	1.00	0.80	0.75	0.45	0.85	0.20	0.15

As shown in Table ??, all true traumatic fall impacts generated $A_{\text{total}} \geq 3.06g$, significantly surpassing the calibrated threshold of $\text{Th}_{\text{impact}} = 2.80g$. In contrast, aggressive daily activities such as rapid sitting and stair descent peaked at $1.70g$ and $1.92g$ respectively. Controlled prayer prostration demonstrated slow kinematic rotation without the preceding $< 0.5g$ freefall signature ($A_{\text{total}} = 0.75g$), confirming that the multi-stage algorithm effectively eliminates false triggers during routine non-emergency physical movement.

In addition, across 25 intentional dropped-device trials, the 10-second acoustic/visual cancellation grace period achieved a 100% false-alarm suppression rate when cancelled by the operator within the allocated window.

4.2 Wireless Telemetry & Failover Latency Analysis

The communication reliability of the primary LoRaWAN (AU915, SF7, 125 kHz) link and secondary Wi-Fi fallback was characterized under simulated industrial noise.

- **LoRaWAN Airtime & Packet Delivery:** The 4-byte binary telemetry payload achieved a total physical airtime of $T_{\text{packet}} = 36.1 \text{ ms}$. Across 500 transmission trials distributed across the facility, the primary LoRa link achieved a packet delivery success rate of 98.4%, with an average uplink latency of $142 \text{ ms} \pm 18 \text{ ms}$.

- **Autonomous Wi-Fi Failover:** Under forced LoRaWAN concentrator disconnection (simulating gateway failure or deep structural blockage), the client node executed 2 unacknowledged retries (2.1 s timeout) before successfully establishing a Wi-Fi socket connection and dispatching the emergency packet. The total end-to-end failover dispatch latency was measured at $3.96 \text{ s} \pm 0.35 \text{ s}$, well within the critical 10-second emergency response SLA.

4.3 BLE Indoor Positioning & Gateway Attenuation Profiling

4.3.1 Gateway RSSI Attenuation Characterization

Empirical signal attenuation profiles were gathered across the facility to benchmark the RF sensitivity and stability of the Minew G1, Mokosmart MKGW3, and Mokosmart LW003-B gateways. Table ?? presents the measured average RSSI (dBm) at stepped distances from 0 m to 7 m.

Table 4.2: Empirical Gateway RSSI Attenuation Across Stepped Distances

Distance (d , meters)	Minew G1 (dBm)	Mokosmart MKGW3 (dBm)	Mokosmart LW003-B (dBm)
0 m (Near Field)	−30 dBm	−60 dBm	−58 dBm
1 m	−50 dBm	−75 dBm	−72 dBm
2 m	−65 dBm	−80 dBm	−78 dBm
3 m	−70 dBm	−85 dBm	−82 dBm
4 m	−70 dBm	−85 dBm	−84 dBm
5 m	−65 dBm	−80 dBm	−80 dBm
6 m	−55 dBm	−75 dBm	−76 dBm
7 m	−50 dBm	−75 dBm	−74 dBm

As shown in Table ??, the Minew G1 exhibited higher receiver sensitivity (−50 dBm at 1 m vs. −75 dBm for Mokosmart). However, at distances beyond 4 m, constructive multipath reflections from metallic structural walls caused the RSSI to artificially rise (reaching −50 dBm at 7 m). This non-monotonic behavior demonstrates the fundamental limitation of raw geometric trilateration in enclosed industrial environments.

4.3.2 Localization Algorithm Accuracy Comparison

To resolve multipath anomalies, four localization methodologies were implemented and benchmarked across a 4-gateway layout (`ble_positioning_4gw.py`). Table ?? summarizes the resulting positioning performance.

The experimental data confirms that while discrete Kalman filtering substantially reduces raw coordinate error (from 3.85 m to 2.15 m), the Fuzzy Inference System mapping to predefined Room Zones achieved superior real-world reliability (94.8% room classification accuracy), ensuring clear location tags in emergency dispatches.

Table 4.3: Indoor Positioning Performance Benchmark Across Localization Methods

Localization Algorithm	Mean Coordinate Error	Room Classification Accuracy
Raw RSSI Trilateration	3.85 m ± 1.42 m	62.4%
Kalman-Filtered Trilateration	2.15 m ± 0.68 m	78.6%
Weighted Centroid (WCL)	2.40 m ± 0.85 m	81.2%
Fuzzy Logic Room Classifier	N/A (Symbolic)	94.8%

4.4 Power Budget Kinetics & Server UPS Reliability

4.4.1 Client Battery Discharge Characterization

Continuous discharge experiments were conducted on the wearable client comparing 600 mAh vs. 1100 mAh LiPo cells under active 100 ms BLE advertising and periodic LoRaWAN keepalive transmissions:

- **600 mAh LiPo Cell:** Initiated at 9:30 AM (4.20 V), the device depleted to hardware cutoff (3.40 V) at 3:15 PM, yielding a total operational runtime of 5 hours 45 minutes. Linear fuel gauge interpolation reported a premature shutdown percentage of 35%.
- **1100 mAh LiPo Cell:** Operating under identical workloads, the 1100 mAh cell achieved > 24.5 hours of continuous operation, providing an industrial Safety Factor (SF) of:

$$SF = \frac{\text{Operational Runtime}}{\text{Standard Shift Duration}} = \frac{24.5 \text{ h}}{12.0 \text{ h}} = \mathbf{2.04} \quad (4.1)$$

- **Non-Linear Voltage Remapping:** Reworking the battery estimation algorithm to follow the true electro-chemical discharge curve (mapping 4.20 V → 100%, 3.70 V → 50%, and 3.30 V → 0%) eliminated premature cutoff reporting, providing monotonic, high-accuracy battery telemetry.

4.4.2 Server UPS Mains Failure Endurance

The central Raspberry Pi 5 server connected to the dual-cell Waveshare UPS HAT was evaluated under simulated total mains power loss:

- **Mains Disconnection Timestamp:** 4:10 PM (Day 1).
- **Automated Low-Battery SMS Warning:** Triggered at 1:34 AM (Day 2) when cell voltage reached 3.30 V.
- **Safe Automated Shutdown:** Executed cleanly by `ups-shutdown.service` at 2:02 AM (Day 2) at 3.20 V (0.0% capacity).
- **Total Server Backup Endurance:** 9 hours 52 minutes of continuous multithreaded server, MQTT broker, and web GUI operation during building power blackout.

4.5 Hardware Diagnostic Analysis: Raspberry Pi 5 RP1 PMIC Lockup

During testing of early edge server assemblies, a critical non-recoverable boot failure occurred on the Raspberry Pi 5 platform. Connecting a UART-to-USB console cable (115200 baud) to the dedicated debug port yielded the boot log:

```

1 0.97 RPi: BOOTSYS release VERSION:2226a853 DATE: 2025/12/08
2 0.98 AON_RESET: 00000003 PM_RSTS 00001000
3 0.99 POWER_OFF_ON_HALT: 0 WAIT_FOR_POWER_BUTTON 0 power-on-reset 1
4 0.99 RP1_BOOT chip ID: 0x0000cf00
5 1.00 I2C error @ 800058a2

```

Root Cause Analysis: The expected RP1 chip identifier is 0x20001927. The incorrect ID (0x0000cf00) combined with the I2C bus error indicated that the external UPS HAT was back-powering and pulling the shared I2C telemetry lines high *before* the Pi 5’s internal PMIC (3.3 V rail) had completed its power-on sequence. This created bus contention that locked the RP1 I/O controller. The issue was permanently mitigated by integrating Schottky isolation diodes on the I2C lines and introducing a 200 ms power-sequencing delay circuit.

4.6 Bill of Materials (BOM) & Unit Financial Analysis

Table ?? outlines the complete Bill of Materials for the fully functional prototype platform.

Table 4.4: Emergency Response System Prototype Procurement & Bill of Materials

Component Description	Qty	Unit Cost (AUD)	Total (AUD)	Supplier
Raspberry Pi Pico 2WH	3	\$14.30	\$42.90	Core Elect
Waveshare LoRaWAN HAT for Pico (SX1262)	4	\$24.15	\$96.60	Core Elect
Seeed Studio XIAO nRF52840 BLE Module	4	\$19.25	\$77.00	Core Elect
InvenSense MPU6050 6-DOF IMU	3	\$4.55	\$13.65	Core Elect
LiPo Battery 1S 3.7V 1100mAh	2	\$8.65	\$17.30	Core Elect
Waveshare Pico UPS Module (Type B)	3	\$21.90	\$65.70	Core Elect
915MHz LoRa Antenna & RF Cable	2	\$20.50	\$41.00	Core Elect
Emergency Push Button & Tactile Switches	4	\$1.30	\$5.20	Core Elect
Raspberry Pi 5 Model B (4GB RAM)	1	\$179.55	\$179.55	Core Elect
Waveshare SX1302/SX1303 LoRa Gateway HAT	1	\$159.95	\$159.95	Core Elect
Waveshare Pi 5 UPS HAT & 18650 Cells	1	\$99.40	\$99.40	Core Elect
Minew G1 BLE-to-Wi-Fi Gateway	1	\$150.00	\$150.00	DigiKey A
Veroboard, Connectors, Wiring & Hardware	1	\$17.85	\$17.85	Core Elect
Total Prototype Hardware Expenditure			\$966.10	<i>Excl. Ship</i>

At scale (> 500 units), migrating to a single consolidated 4-layer PCB with SMT components is projected to reduce the per-wearable hardware BOM from \$87.50 AUD to under \$32.00 AUD, establishing a highly competitive price-to-performance ratio.

5. Discussion & Commercialisation Strategy

5.1 Analysis of Engineering Insights & Trade-Offs

5.1.1 Wireless Multi-Protocol Redundancy Synergy

The empirical results presented in Chapter ?? validate the architectural premise that no single wireless communication protocol satisfies all industrial safety constraints [9]. While 2.4 GHz Wi-Fi provides continuous session persistence and high data throughput, its high power consumption (> 100 mA) and severe attenuation through structural steel make it unsuitable as a sole emergency uplink. Conversely, sub-GHz LoRaWAN (AU915) offers unmatched penetration through heavy machinery and metallic vehicle chassis, but its low bandwidth and ALOHA duty-cycle restrictions preclude continuous real-time streaming.

By establishing LoRaWAN as the primary low-power telemetry backbone with an automated, state-machine-driven failover to Wi-Fi upon packet acknowledgement timeout, the ERS achieves a dual-redundant communication framework. The 3.96 second failover latency guarantees that in scenarios where a local LoRa concentrator is compromised or an RF channel suffers localized jamming, critical distress signals successfully reach the server via secondary LAN infrastructure.

5.1.2 Spatial Localization: The Coordinate vs. Room Zone Paradigm

A major engineering finding of this project is the practical limitation of continuous (x, y) coordinate trilateration in cluttered industrial environments. As documented in Section ??, severe multipath reflections from metal sheet walls introduce non-monotonic RSSI gradients, causing raw trilateration algorithms to experience coordinate jumping errors exceeding 3.8 meters. While discrete Kalman filtering smooths transient jitter, it cannot completely resolve structural shadowing.

Consequently, the strategic shift toward symbolic, discrete **Room Zone classification** supported by Mamdani Fuzzy Inference represents a major operational breakthrough. In mission-critical emergency dispatch, responders do not require floating-point Cartesian coordinates; rather, actionable linguistic location data (e.g., “Office 1”, “Main Assembly Bay”, “Spray Booth”) is required to direct immediate physical intervention. Achieving a 94.8% room classification accuracy validates this engineering trade-off.

5.1.3 Economic Viability: CAPEX vs. OPEX Subscription Models

Commercial lone-worker solutions (such as Blackline Safety and SoloProtect) impose mandatory monthly recurring SaaS fees ranging from \$30 to \$60 AUD per device [3]. For an industrial facility employing 20 workers, recurring operational expenditures exceed \$7,200 – \$14,400 AUD annually, representing a significant barrier for SMEs.

In contrast, the Micromax ERS operates on a self-hosted, on-premises edge computing model. As established in the procurement analysis, a 20-node deployment carries an initial capital equipment cost of approximately \$2,200 AUD (amortized across gateways, server, and client

nodes), with zero recurring software licensing fees. This delivers over an 80% reduction in Total Cost of Ownership (TCO) over a three-year operating lifecycle, while ensuring complete organizational data privacy and on-premises audit compliance.

5.2 Environmental & Technical Limitations

Despite the successful demonstration of the functional prototype, several technical constraints were identified during environmental testing:

1. **LoRaWAN Concentrator Half-Duplex Constraints:** Standard industrial LoRa gateways (such as the SX1302 HAT) are half-duplex. When transmitting downlink acknowledgement packets (Class A RX1/RX2 windows), the concentrator cannot simultaneously receive uplinks from other nodes. In high-density emergency scenarios where multiple workers trigger simultaneous alarms, uncoordinated downlinks could increase uplink packet collision rates.
2. **BLE Signal Variance in Dynamic Environments:** Real-time physical movement of personnel, forklifts, and large metallic objects within the facility introduces dynamic RSSI fluctuations (± 6 dBm). While fuzzy filtering mitigates boundary ambiguity, transition zones between adjacent open-plan bays remain sensitive to environmental noise.
3. **RP2040 Sleep Current Optimization:** In the current veroboard prototype, the RP2040 operates in active execution mode with periodic light-sleep states. Future commercial firmware must implement deep dormant states with hardware interrupt waking to extend the 1100 mAh battery lifespan from 24.5 hours to several weeks between charges.

5.3 60-Day Commercialisation Roadmap

Transitioning the Emergency Response System from the current lab-validated, functional prototype into a pilot-ready commercial product forms the core objective of the commercialisation phase (structured over a 60-day industrial productization lifecycle):

1. **Hardware Consolidation (Custom 4-Layer PCB Design):**
 - Replace the multi-board veroboard assembly (Pico + LoRa HAT + XIAO BLE + IMU) with a single, monolithic 4-layer Surface-Mount Technology (SMT) printed circuit board.
 - Integrate the RP2040 microcontroller directly alongside the Semtech SX1262 transceiver, Nordic nRF52840 SoC, MPU6050 (or high-accuracy LSM6DSOX), dynamic I2C fuel gauge, and an onboard USB Type-C battery charging controller.
 - Optimize RF trace routing, ground plane isolation, and matched $50\ \Omega$ impedance transmission lines for both the 915 MHz LoRa and 2.4 GHz BLE ceramic patch antennas.
2. **Mechanical Enclosure Design & Tooling:**

- Design an ergonomic, ruggedized injection-molded enclosure fabricated from high-impact polycarbonate (PC/ABS) with an integrated silicone gasket, achieving an IP67 water and dust ingress protection rating.
- Incorporate versatile industrial mounting accessories: heavy-duty spring-loaded belt clip, breakaway safety lanyard eyelet, and high-visibility emergency push-button with tactile shroud to prevent accidental actuation.

3. Software Hardening & Enterprise Integration:

- Containerize the edge server backend using Docker to ensure seamless, single-command on-premises deployment.
- Implement role-based access control (RBAC), TLS/SSL encryption across all internal MQTT/HTTP traffic, and RESTful API webhooks for integration with enterprise HR and access control systems (Active Directory, Gallagher, etc.).
- Implement Over-the-Air (OTA) firmware update capabilities across the client fleet via secondary Wi-Fi connections.

4. Pilot Deployment & Manufacturing Handover:

- Execute a 30-day live customer pilot deployment at representative industrial sites (e.g., Goldstream RV manufacturing plant), tracking uptime, false-alarm frequency, and operator feedback.
- Prepare Design for Manufacturing (DFM) and Design for Test (DFT) documentation, automated bed-of-nails PCB test jigs, and regulatory compliance dossiers for Australian RCM (AS/NZS CISPR 32) certification.

5. Strategic Drone Technology Synergy:

- Conduct feasibility analysis on integrating autonomous unmanned aerial vehicles (UAVs / drones) with the ERS backend.
- In large-scale outdoor industrial yards, agricultural holdings, or open manufacturing storage areas, an active emergency trigger from an ERS client node can autonomously dispatch a tethered/docked drone to the estimated GPS/zone coordinate, streaming live optical and thermal video directly to emergency response teams before first responders physically arrive on site.

6. Conclusion & Future Work

6.1 Summary of Engineering Achievements

This technical report has presented the comprehensive systems engineering design, empirical characterization, and commercialisation roadmap of the Micromax Emergency Response System (ERS) for industrial organisations. Evolving from an exploratory university capstone proof-of-concept (ECTE351) to an integrated, field-tested functional industrial prototype (ECTE399 R&D), the project successfully addresses critical Work Health and Safety (WHS) compliance mandates for lone and isolated workers.

The major technical milestones and engineering deliverables achieved in this work include:

1. **Deterministic Multi-Modal Emergency Detection:** Implemented robust embedded logic on the wearable client distinguishing between deliberate Personal Emergencies (> 2.0 s press-and-hold) and facility-wide Mass Emergencies (5 rapid presses within 3.0 s), alongside resolving critical GPIO15 hardware reset conflicts.
2. **Kinematic Fall Detection with Grace Cancellation:** Calibrated a 6-DOF IMU algorithm capable of reliably capturing freefall ($< 0.5g$) and high-impact ($> 2.8g$) dynamics while rejecting routine Activities of Daily Living (e.g., prostration in prayer, sitting, stair descent), reinforced by a 10-second audible cancellation window that achieved a 100% false-alarm suppression rate in testing.
3. **Dual-Redundant Wireless Telemetry Architecture:** Developed a fail-safe communication pipeline combining primary sub-GHz LoRaWAN (AU915, SF7) utilizing an ultra-compact 4-byte binary payload (36.1 ms airtime, 98.4% delivery rate) with an autonomous 3.96 second failover to IEEE 802.11 Wi-Fi upon transmission timeout.
4. **Symbolic Indoor Positioning via Room Zone Mapping:** Deployed a multi-gateway BLE tracking infrastructure and established that Mamdani Fuzzy Inference mapping to polygon-based Room Zones outperforms continuous coordinate trilateration in metallic multipath environments, achieving a 94.8% room classification accuracy.
5. **Centralized Edge Processing Automated Multi-Channel Dispatch:** Built an on-premises Raspberry Pi 5 server hosting multithreaded ingress daemons, an SQLite relational database, dynamic web-based floor plan configuration, and parallel dispatch workers (cellular SMS via Twilio, SMTP email, acoustic sirens, and optocoupled dry-contact industrial relays).
6. **Power UPS Resilience:** Validated a 24.5 hour wearable battery lifespan (1100 mAh LiPo, SF = 2.04 for 12 h shifts) and demonstrated 9 hours 52 minutes of autonomous server survivability during total mains power blackout with automated SMS warnings and safe shutdown at 3.2 V.

6.2 Future Research & Engineering Directions

To build upon the foundation established in this report and support full-scale commercial deployment, several strategic engineering directions are identified:

- **Bluetooth 5.1 Angle of Arrival (AoA) Integration:** Partnering with commercial RTLS providers (such as BlueIOT) to evaluate multi-antenna AoA array locator gateways. By measuring RF phase differences, AoA promises sub-meter (< 0.5 m) 3D positioning accuracy, enabling precise tracking within dense machinery bays and multi-tier industrial racking.
- **Edge-AI / TinyML Kinematic Classification:** Deploying lightweight neural network models (e.g., 1D Convolutional Neural Networks or TinyML Random Forests) directly onto the ARM Cortex-M4F core using TensorFlow Lite for Microcontrollers. This will enable complex temporal motion gesture classification, eliminating fixed threshold sensitivity across diverse human body masses and operational postures.
- **Ultra-Low-Power Dormant State Optimization:** Implementing deep-sleep power gating with accelerometer interrupt waking (μ A-level quiescent draw) to extend client battery endurance from days to several months per charge cycle.
- **Autonomous Drone-Assisted Emergency Verification:** Developing software hooks within the central server to autonomously task docked industrial UAVs to outdoor emergency coordinates, providing instant live video reconnaissance to incident controllers during critical search-and-rescue operations.

References

- [1] *Managing the work environment and facilities code of practice*, Section 4: Remote or Isolated Work, Safe Work Australia, 2020. [Online]. Available: <https://www.safeworkaustralia.gov.au/doc/model-code-practice-managing-work-environment-and-facilities>
- [2] *Occupational health and safety management systems – requirements with guidance for use*, Geneva, Switzerland: International Organization for Standardization, 2018.
- [3] S. Memon, F. M. Kundi, A. Keerio, and R. Memon, “Design and implementation of smart safety devices for industrial lone workers,” *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 8, pp. 450–458, 2022. DOI: [10.14569/IJACSA.2022.0130852](https://doi.org/10.14569/IJACSA.2022.0130852)
- [4] A. K. Bourke, J. V. O’Brien, and G. M. Lyons, “Evaluation of a threshold-based tri-axial accelerometer fall detection algorithm,” *Gait & Posture*, vol. 26, no. 2, pp. 194–199, 2007. DOI: [10.1016/gaitpost.2006.09.012](https://doi.org/10.1016/gaitpost.2006.09.012)
- [5] F. Zafari, A. Gkelias, and K. K. Leung, “A survey of indoor localization systems and technologies,” *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2568–2599, 2019. DOI: [10.1109/COMST.2019.2911558](https://doi.org/10.1109/COMST.2019.2911558)
- [6] S. Subedi and J.-Y. Pyun, “Practical RSSI-based indoor localization using fingerprinting and trilateration,” in *Proceedings of the International Conference on Information Networking (ICOIN)*, 2017, pp. 723–725. DOI: [10.1109/ICOIN.2017.7899522](https://doi.org/10.1109/ICOIN.2017.7899522)
- [7] S. S. Chawathe, “Low-latency indoor localization using Bluetooth Low Energy beacons and Kalman filters,” *IEEE Transactions on Mobile Computing*, vol. 20, no. 4, pp. 1412–1424, 2021. DOI: [10.1109/TMC.2019.2959821](https://doi.org/10.1109/TMC.2019.2959821)
- [8] A. Yassin, Y. Nasser, M. Awad, M. Jansen, and S. Boudjit, “Recent advances in indoor localization: A survey on RF and non-RF technologies,” *IEEE Communications Surveys & Tutorials*, vol. 19, no. 2, pp. 1327–1376, 2017. DOI: [10.1109/COMST.2016.2632427](https://doi.org/10.1109/COMST.2016.2632427)
- [9] LoRa Alliance and Wireless Broadband Alliance, “Wi-fi & LoRaWAN deployment synergies: Expanding business opportunities with combined technologies,” *White Paper*, 2020. [Online]. Available: https://lora-alliance.org/resource_hub/wi-fi-lorawan-deployment-synergies/
- [10] Raspberry Pi Ltd, *RP2040 microcontroller datasheet: A high performance microcontroller with flexible digital interfaces*, 2023. [Online]. Available: <https://datasheets.raspberrypi.com/rp2040/rp2040-datasheet.pdf>
- [11] InvenSense Inc., “MPU-6000 and MPU-6050 product specification revision 3.4,” *Document Number: PS-MPU-6000A-00*, 2013.
- [12] Raspberry Pi Ltd, *Raspberry Pi 5 product brief*, 2023. [Online]. Available: <https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-product-brief.pdf>

- [13] Semtech Corporation, “SX1302 long range corecell gateway baseband processor datasheet,” *Semtech Technical Documentation*, no. DS.SX1302.W.APP, 2020.

A. Core Firmware & Server Code Listings

This appendix provides key production source code listings implemented during the functional prototype development phase.

A.1 Client Communication & Dual-Failover Logic (ClientDeviceV2.py)

Listing A.1: Pico Client Multi-Modal Failover State Machine

```
1 import time, struct, machine, network, socket
2 from machine import Pin, SPI, I2C
3
4 class ClientDevice:
5     def __init__(self, dev_id=0x01):
6         self.device_id = dev_id
7         self.seq_num = 0
8         self.lora_retries_max = 2
9         # Initialize isolated hardware pins (Resolving GPIO15 conflict)
10        self.btn = Pin(15, Pin.IN, Pin.PULL_UP)
11        self.lora_reset = Pin(13, Pin.OUT)
12        self.buzzer = Pin(14, Pin.OUT)
13        self.wlan = network.WLAN(network.STA_IF)
14
15    def pack_payload(self, event_type, battery_pct):
16        # 4-Byte compact encoding: DevID (1B), Event/Batt (1B), SeqNum (2B)
17        batt_code = int((battery_pct / 100.0) * 15) & 0x0F
18        event_code = (event_type & 0x0F) << 4
19        byte1 = event_code | batt_code
20        payload = struct.pack(">BBH", self.device_id, byte1, self.
21                               seq_num)
22        self.seq_num = (self.seq_num + 1) % 65536
23        return payload
24
25    def send_emergency(self, event_type, battery_pct):
26        payload = self.pack_payload(event_type, battery_pct)
27        lora_success = False
28
29        # Attempt primary transmission via LoRaWAN
30        for attempt in range(self.lora_retries_max):
31            if self.transmit_lorawan(payload):
32                lora_success = True
33                print(f"[COMMS] LoRaWAN transmission success on attempt {attempt+1}")
34                break
35
36        time.sleep_ms(200)
```

```

35
36     # Autonomous failover to Wi-Fi if LoRaWAN is unavailable
37     if not lora_success:
38         print("[WARN] LoRaWAN unacknowledged. Triggering Wi-Fi failover...")
39         self.transmit_wifi_failover(payload)
40
41     def transmit_wifi_failover(self, payload):
42         self.wlan.active(True)
43         self.wlan.connect("Micromax_Safety_Net", "EmergencyKey2026")
44         timeout = 50
45         while not self.wlan.isconnected() and timeout > 0:
46             time.sleep_ms(100)
47             timeout -= 1
48
49         if self.wlan.isconnected():
50             s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
51             s.sendto(payload, ("192.168.115.49", 5005))
52             s.close()
53             print("[COMMS] Wi-Fi UDP emergency packet dispatched successfully.")

```

A.2 Discrete Kalman Filter RSSI Processor (ble_positioning_4gw.py)

Listing A.2: Discrete 1D Kalman Filter Implementation for Gateway RSSI

```

1 class KalmanRSSI:
2     def __init__(self, process_noise_q=0.008, measurement_noise_r=0.65):
3         :
4         self.q = process_noise_q # Process variance
5         self.r = measurement_noise_r # Measurement noise
6         self.x = -65.0 # Estimated state
7         self.p = 1.0 # Estimation error covariance
8         self.k = 0.0 # Kalman gain
9
10    def update(self, measurement):
11        # Time Update (Predict)
12        p_pred = self.p + self.q
13        # Measurement Update (Correct)
14        self.k = p_pred / (p_pred + self.r)
15        self.x = self.x + self.k * (measurement - self.x)
16        self.p = (1.0 - self.k) * p_pred
17        return self.x

```

B. Supplementary Engineering Data & Hardware Schematics

This appendix provides supplementary hardware configuration parameters, gateway network mappings, and database DDL schemas.

B.1 BLE Gateway Network Topology & MAC Allocations

Table ?? details the network static IP and MAC address allocations configured on the Micromax facility testbed.

Table B.1: BLE Gateway Network Topology Configurations

Gateway Node	MAC Address	Static IP Address	Physical Placement / Room
Minew G1 – GW1	AC:23:3F:C2:59:32	192.168.115.240	Main Engineering Office (Ceiling Anchor 1)
Minew G1 – GW2	AC:23:3F:C2:6E:CB	192.168.115.241	Boardroom / Meeting Space (Anchor 2)
Minew G1 – GW3	AC:23:3F:C2:6E:CC	192.168.115.242	Kitchen / Breakout Area (Anchor 3)
Mokosmart MKGW3	D4:36:39:1A:45:B1	192.168.115.243	R&D Prototype Testing Bay (PoE Anchor 4)
Edge Server Broker	B8:27:EB:A4:91:20	192.168.115.49	Central Server Rack (Port 1883 / 5000)

B.2 Client Microcontroller Hardware Pinout Mapping

Table ?? outlines the physical GPIO pin allocations for the Raspberry Pi Pico 2WH wearable client node.

B.3 SQLite Database Relational Schema DDL

Listing B.1: Relational SQLite Database Table Definitions

```

1 CREATE TABLE staff (
2     staff_id INTEGER PRIMARY KEY AUTOINCREMENT,
3     full_name TEXT NOT NULL,
4     email TEXT NOT NULL,

```

Table B.2: Raspberry Pi Pico 2WH Wearable Client Pinout Configuration

Pico Pin / GPIO	Direction	Peripheral / Signal	Function / Interface Description
GPIO4	Bi-directional	MPU6050 I2C SDA	Sensor data bus (400 kHz Fast Mode)
GPIO5	Output	MPU6050 I2C SCL	Sensor clock line
GPIO10	Output	SX1262 SPI SCK	LoRaWAN radio SPI clock
GPIO11	Output	SX1262 SPI MOSI	LoRaWAN radio SPI Master-Out Slave-In
GPIO12	Input	SX1262 SPI MISO	LoRaWAN radio SPI Master-In Slave-Out
GPIO13	Output	SX1262 NRESET	Dedicated hardware radio reset line
GPIO14	Output	Buzzer / Status LED	Piezoelectric acoustic warning (85 dB)
GPIO15	Input	Emergency Button	Tactile distress trigger with 100nF RC filter
GPIO16	Output	SX1262 NSS (CS)	LoRaWAN radio SPI Chip Select (Active Low)
GPIO20	Input	SX1262 DIO1	LoRa radio interrupt (Tx/Rx Done)
ADC0 (GPIO26)	Analog Input	UPS Fuel Gauge	Battery cell analog voltage divider
VSYS / GND	Power Rail	LiPo Power Input	3.7 V nominal from UPS Boost Converter

```
5     phone_number TEXT NOT NULL,
6     department TEXT,
7     special_medical_notes TEXT,
8     is_active INTEGER DEFAULT 1
9 );
10
11 CREATE TABLE devices (
12     device_id INTEGER PRIMARY KEY,
13     lora_deveui TEXT UNIQUE,
14     ble_mac_address TEXT UNIQUE,
15     assigned_staff_id INTEGER,
16     battery_percentage REAL,
17     last_heartbeat DATETIME,
18     FOREIGN KEY(assigned_staff_id) REFERENCES staff(staff_id)
19 );
20
21 CREATE TABLE room_zones (
22     zone_id INTEGER PRIMARY KEY AUTOINCREMENT,
23     zone_name TEXT NOT NULL,
24     x_min REAL, y_min REAL,
25     x_max REAL, y_max REAL,
26     floor_level INTEGER DEFAULT 1
27 );
28
29 CREATE TABLE incident_logs (
30     log_id INTEGER PRIMARY KEY AUTOINCREMENT,
31     timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
32     device_id INTEGER,
33     event_type TEXT,
34     raw_rssi_data TEXT,
35     estimated_x REAL, estimated_y REAL,
36     resolved_room_name TEXT,
37     dispatch_status TEXT,
38     is_drill INTEGER DEFAULT 0,
39     FOREIGN KEY(device_id) REFERENCES devices(device_id)
40 );
```